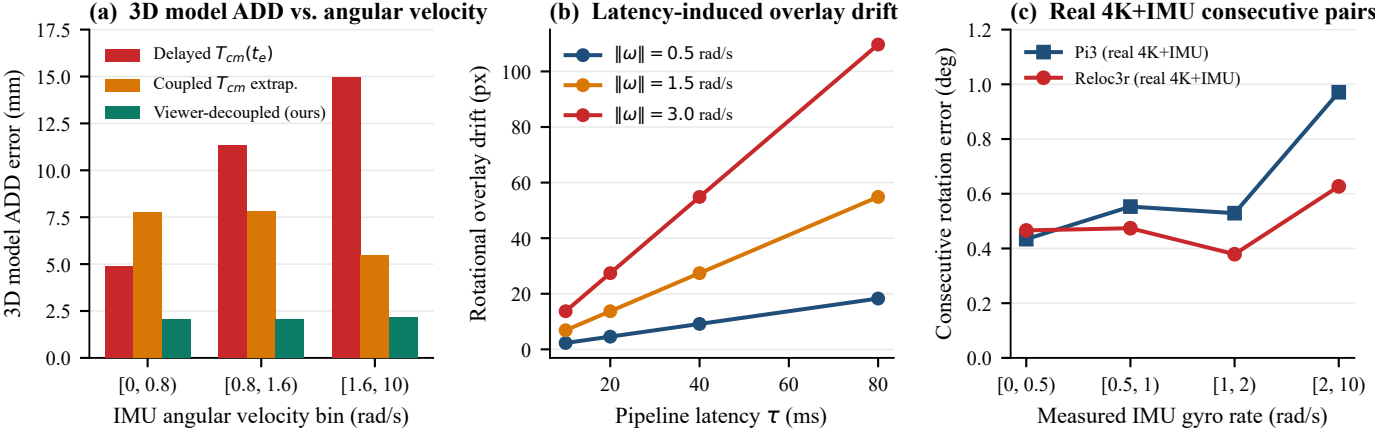

(a) 3D model ADD vs. angular velocity
3D model ADD error (mm)
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
Delayed $T_{cm}(t_e)$
Coupled T_{cm} extrap.
Viewer-decoupled (ours)
[0, 0.8)
[0.8, 1.6)
[1.6, 10)
IMU angular velocity bin (rad/s)
(b) Latency-induced overlay drift
Rotational overlay drift (px)
100
80
60
40
20
0
$\|\omega\|$ = 0.5 rad/s
$\|\omega\|$ = 1.5 rad/s
$\|\omega\|$ = 3.0 rad/s
20
40
60
80
Pipeline latency τ (ms)
(c) Real 4K+IMU consecutive pairs
Consecutive rotation error (deg)
1.2
1.0
0.8
0.6
0.4
0.2
0.0
Pi3 (real 4K+IMU)
Reloc3r (real 4K+IMU)
[0, 0.5)
[0.5, 1)
[1, 2)
[2, 10)
Measured IMU gyro rate (rad/s)